

# DIKSHIT CONSULTANTS & ENGINEERS PVT. LTD.

## 13. INSULATION

### FOR G.I. DUCTING - SPECIFICATION FOR THERMAL & ACOUSTIC INSULATION

#### SCOPE

The scope of this section comprises the supply and application of insulation conforming to these specifications.

#### 13.1. DUCT THERMAL INSULATION

##### MATERIAL

1. Insulation material shall be Closed Cell Elastomeric **NITRILE RUBBER**
2. Density of Material shall be between 40 to 60 Kg/m<sup>3</sup>
3. Thermal conductivity of elastomeric NITRILE RUBBER shall not exceed 0.034 W/m.K at mean temperature of 0°C
4. Insulation material shall have anti-microbial product, which is EPA (Environmental Protection Agency), USA approved, as an integral part of insulation that cannot be washed off or worn off. **(NITRILE RUBBER is anti-Microbial polymer)**
5. It shall give enhanced level of protection against harmful Microbes such as bacteria, mold, mildew and fungi and should confirm to following standards: Fungi Resistance – ASTM G21 and Bacterial resistance – ASTM G 22 / ASTM 2180.
6. Insulation shall have fire performance such that it passes **ASTM-E-84 test** Method
7. Material should be FM (Factory Mutual), USA approved.
8. Water vapor permeability shall not exceed  $1.74 \times 10^{-14}$  Kg / (m.s.Pa), i.e. Moisture Diffusion Resistance Factor or 'μ' value should be minimum 10000.

**Thickness of the insulation shall be as specified for the individual application.**

#### **THICKNESS SELECTION CHART FOR NITRILE RUBBER INSULATION**

Design Basis: Condensation Control

#### **OUTDOOR ,ATTIC & UNCONDITIONED SPACES VENTED TO OUTDOOR**

Design Conditions: 32 Deg. C & 85% RH (as per ASHRAE guidelines)

#### **DUCT INSULATION**

| Duct                                        | Required Thickness (mm) |
|---------------------------------------------|-------------------------|
| Supply Air Duct (Line Temperature 14 Deg.C) | 25                      |
| Return Air Duct (Line Temperature 22 Deg.C) | 25                      |

#### **THICKNESS SELECTION CHART FOR NITRILE RUBBER INSULATION**

Design Basis: Condensation Control

#### **INDOOR (CONDITIONED / SEMICONDITIONED AREAS)**

Design Conditions: 26.8 Deg.C & 85% RH (With Proper Ventilation)

#### **DUCT INSULATION**

| Duct                                                     | Required Thickness (mm) |
|----------------------------------------------------------|-------------------------|
| Supply Air Duct (Line Temperature 14 Deg.C)              | 19                      |
| Return Air Duct (Line Temperature 22 Deg.C)              | 13                      |
| Supply Air Duct in Return Air Path (Line Temp. 14 Deg.C) | 13                      |

Noted



| <b>THICKNESS SELECTION CHART FOR NITRILE RUBBER INSULATION</b> |                                        |
|----------------------------------------------------------------|----------------------------------------|
| Design Basis: Condensation Control                             |                                        |
| <b>Non - Coastal Areas</b>                                     | Pune, Delhi, Hyderabad, Bangalore etc. |
| <b>DUCT INSULATION</b>                                         |                                        |
| <b>Duct</b>                                                    | <b>Required Thickness (mm)</b>         |
| Supply Air Duct (Line Temperature 14 Deg.C)                    | 19                                     |
| Return Air Duct (Line Temperature 22 Deg.C)                    | 13                                     |
| Supply Air Duct in Return Air Path (Line Temp. 14 Deg.C)       | 13                                     |

External thermal insulation shall be provided as follow:

The thickness of the NITRILE RUBBER shall be as shown on drawing or identified in the schedule of quantity. Following installation procedure should be adopted:

1. Duct surfaces shall be cleaned to remove all grease, oil, dirt, etc. prior to carrying out insulation work.
2. Measurement of surface dimensions shall be taken properly to cut closed cell elastomeric rubbers sheets to size with sufficient allowance in dimension.
3. Material shall be fitted under compression and no stretching of material should be allowed.
4. A thin film of adhesive shall be applied on the back of the insulating material sheet and then on to the metal surface.
5. When adhesive is tack dry, insulating material sheet shall be placed in position and pressed firmly to achieve a good bond.
6. All longitudinal and transverse joints shall be sealed as per manufacturer recommendations.
7. The adhesive shall be strictly as recommended by the manufacturer.
8. Detailed Application specifications are as per the manufacturer's recommendation.

Noted

### **13.2. INSTALLATION OF DUCTS EXPOSED DIRECTLY TO SUNLIGHT:**

For installations exposed to sunlight, after giving 36 hours curing time for the adhesive apply manufacturer's recommended UV/Mechanical Protection. Please refer the separate detailed guidelines on UV/Mechanical Protection.

### **13.3. INSULATION FOR CHILLED WATER/DRAIN PIPING**

The complete chilled water piping including with pipes, bends, flanges, valves, hanger support etc. will be insulated as per the following -

#### **MATERIAL**

1. Insulation material shall be Closed Cell Elastomeric Nitrile Rubber
2. Density of Material shall be between 40 to 60 Kg/m<sup>3</sup>
3. Thermal conductivity of elastomeric nitrile rubber shall not exceed 0.035 W/m<sup>2</sup>K at an average temperature of 0°C
4. The insulation shall have fire performance such that it passes Class 1 as per BS476 Part 7 for surface spread of flame as per BS 476 and also pass Fire Propagation requirement as per BS476 Part 6 to meet the Class 'O' Fire category as per 1991 Building Regulations (England & Wales) and the Building Standards (Scotland) Regulations 1990
5. Material should be FM (Factory Mutual), USA approved.
6. Thickness of the insulation shall be as specified for the individual application.

Comply

